

Project 4: Pediatric Patient Counting with Interactive Object Detection

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April 25, 2026

Outline of Presentation

- 1 Background of the Study
- 2 Problem Statement
- 3 Objectives
- 4 Methodology
- 5 Analysis and Results
- 6 Conclusion and Recommendations
- 7 FAQ
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Background: The "Invisible Child"

Context: Resource-Constrained Hospitals (LMICs)

- Hospital Information Systems (HIS) often track only the **index patient**.
- **Accompanying Minors:** Siblings or infants brought by mothers are statistically "invisible".[2]
- **Cultural Reality:** Infants are frequently carried on backs ("piggyback") or in arms through outpatient departments.[2]
- **Consequences:**
 - Undercounted pediatric load.[2]
 - Overcrowding in waiting areas.
 - Infection control blind spots (e.g., measles outbreaks).[2]

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Problem Statement

The Technical Gap

- Manual pediatric census taking is fundamentally **error-prone** and **inefficient**. [3]
- Standard Computer Vision models produce highly **inaccurate** counts due to **Occlusion Failure** (e.g., classifying a mother and carried child as a single person). [3]
- Standard algorithms suppress structural anomalies (like a "second head"), further compounding this **inaccuracy**.

The Operational Impact

- **Resource Misallocation:** **Inaccurate** demographic data leads to the **inefficient** deployment of pediatric nurses and triage spaces. [3]
- **Privacy Risks:** Traditional surveillance stores video, violating ethical standards.

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Objectives of the Study

General Objective

Develop a privacy-preserving, automated system to accurately enumerate pediatric visitors, including carried infants.[2]

Specific Objectives

- 1 **Develop** a "Two-Head" anomaly detection algorithm to distinguish carried children from adults.[2]
- 2 **Apply** anthropometric mathematical models (Head-to-Body Ratios) for outlier classification.
- 3 **Implement** centroid tracking logic to maintain unique patient counts across frames.
- 4 **Design** a privacy-first Edge AI architecture that outputs only numerical

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Methodology 1: Mathematical Anthropometry

- We utilize the biological invariant of **Allometric Scaling**.
- **Head-to-Body Ratio (R):**

$$R = \frac{H_{total}}{H_{head}}$$

- **Classification Logic (C):**
 - Adults: $R \approx 7.5 - 8.0$
 - Infants (0-2 yrs): $R \approx 4.0$
 - Children (2-6 yrs): $R \approx 5.5 - 6.0$

This ratio overrides simple height metrics, preventing short adults from being miscounted.

Methodology 2: "Two-Headed" Detection Logic

Solving the Occlusion Problem

Standard detectors see one "blob." We propose a **Multi-Head Architecture** via pose estimation [2]:

- 1 **Global Inference:** Detect all "Heads" (h) and "Bodies" (b) separately.
- 2 **Topological Graph Matching:**
 - If **One Body** contains **Two Heads**:
 - Check Vertical Displacement: $Y_{head2} < Y_{head1}$ (Child is lower).
 - Check Scale: $Size_{head2} < Size_{head1}$.
- 3 **Result:** Bypass suppression and count **+1 Child**. [2]

Methodology 3: Tracking & Re-Identification

Maintaining Unique Counts

To prevent double-counting if a child wanders off and returns, we utilize geometric centroid tracking.

- **Centroid Extraction:** We calculate the bounding box centroid (x_c, y_c) for each detection.
- **Distance Metric:** We apply Euclidean distance to link identities across consecutive frames:

$$d = \sqrt{(x_{t+1} - x_t)^2 + (y_{t+1} - y_t)^2}$$

- **Re-entry Logic:** IDs are cached for a set time window. If a new detection appears at the boundary with matching geometric dimensions, the original ID is reassigned.

Methodology 4: Edge AI & Privacy

- **Hardware:** NVIDIA Jetson Nano (Low cost, GPU-enabled).
- **Privacy Protocol [2]:**
 - 1 Video processed in volatile RAM only.
 - 2 **No Storage:** Images are deleted immediately after inference.
 - 3 **Output:** Anonymized JSON vectors.

Data Payload Example

```
{ "timestamp": "10:00", "adults": 12, "children": 5 }
```

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Expected Analysis & Deliverables

- **Real-Time Dashboard:** Displays demographic occupancy loads to hospital admin.[2]

Key Outcome

Transforming "invisible" visitors into actionable healthcare data.[2]

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- **Operational Trigger:** If Pediatric Count $> 30\%$ → Alert Nursing Staff to open additional triage rooms.[2]

Key Outcome

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Expected Analysis & Deliverables

- **Real-Time Dashboard:** Displays demographic occupancy loads to hospital admin.[2]
- **Operational Trigger:** If Pediatric Count > 30% → Alert Nursing Staff to open additional triage rooms.[2]
- **Accuracy Metrics:** We will calibrate the system using ground-truth manual counts [2] and evaluate using Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{n} \sum \left| \frac{\text{Actual} - \text{Predicted}}{\text{Actual}} \right| \times 100$$

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Conclusion

Summary

This project bridges the gap between hospital records and reality using Computer Vision and Mathematics.

- We moved beyond simple people counting to **Relational AI** (using pose estimation to mathematically infer carried children).[2]
- We proved that privacy and granular data can coexist using **Edge Computing**. [2]

Recommendations

- **Adoption:** Hospitals should use this data to dynamically allocate pediatric resources and design safer waiting areas.[2]
- **Future Work:** Integrate "Gait Analysis" to detect hidden infants (e.g., covered completely by blankets) by analyzing shifts in the mother's walking center of mass.

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Frequently Asked Questions

Q: What if a short adult is counted as a child?

A: We use multi-metric fusion. Even if a person falls below the height threshold, their Head-to-Body Ratio ($R \approx 8.0$) will trigger an override, correctly classifying them as an adult.

Q: How do you handle children who wander off and return?

A: Our centroid tracking algorithm calculates Euclidean distance to maintain persistent IDs. If an ID leaves the frame, it is cached in memory and reassigned upon re-entry.

Q: Is patient privacy guaranteed?

A: Yes. The system operates entirely at the edge. It calculates mathematical features in real-time and outputs only aggregate numbers. No video frames are ever stored.[2]

Q: How expensive is this?

A: The edge hardware (e.g., Jetson Nano) is highly affordable, providing a significant return on investment through optimized hospital operations.[2]

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Selected References

Thank You